The Canine Rosetta Stone

How Far Are We from Translating What Dogs Say?

A Survey of Cross-Species Communication Decoding
in the Age of Foundation Models

The `woof-to-word` Project

An open, evidence-bounded literature survey

`github.com/mcxiaoxiao/woof2word`

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Abstract

Can a machine turn a bark into a sentence? Consumer products already claim to "translate" dog vocalizations with accuracies above 80%, while landmark projects such as Project CETI, Google's DolphinGemma, and the Earth Species Project apply foundation models to sperm whales, dolphins, and songbirds. This survey asks a deliberately narrow question: *how far are we, really, from translating what a dog says?* We organize the answer around a five-rung "translation ladder"—signal detection, affect classification, intent and context inference, referential semantics, and open-ended dialogue—and argue that today's systems sit firmly on the second rung, with scattered footholds on the third. We synthesize three bodies of evidence: (i) the behavioral biology of canine communication (barks, growls, DogFACS facial actions, body posture, and soundboard button use); (ii) the technical pipeline that would be required for genuine translation (self-supervised acoustic encoding, sequence-structure discovery, semantic grounding, and interactive validation); and (iii) the commercial landscape and its evaluation crisis. We identify a central *semantic gap*: dogs possess a rich affective and referential signaling system but lack the recursive, generative syntax that human translation presupposes, so "translation" is partly a category error. We further show that the field's headline accuracies are largely *unfalsifiable*, because no ground-truth "intended message" exists to score against. We close with a capability roadmap, an honest distance estimate, and an ethics framework for responsible interspecies AI. Our conclusion is neither dismissive nor utopian: useful, validated *affect-and-intent* translation is plausible within this decade, while faithful open-ended translation of canine "language" may remain permanently out of reach—not for lack of model scale, but for lack of a language to translate.

Keywords: animal communication, bioacoustics, foundation models, canine cognition, DogFACS, semantic grounding, interspecies interaction, evaluation, research ethics.

A note on scope and method. This is a narrative, evidence-bounded survey, not a PRISMA systematic review. Sources were gathered through structured web and literature search in November 2026; peer-reviewed publications, preprints, project reports, and journalism are distinguished throughout, and commercial accuracy claims are reported *as claims*, not as verified results. Section 7 treats the verification problem explicitly.

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1 Introduction: The Oldest Dream of Domestication

For at least fifteen thousand years—and by some accounts more than thirty thousand—humans have lived alongside dogs, and for nearly as long we have wanted to know what they are saying. The wish is older than science: folktales grant the hero a magic fern-seed or a drop of dragon's blood that lets him understand the speech of beasts. What is new is that the wish now arrives packaged as a product. A smartphone application records a bark and returns a sentence—"I want to play," "my tummy hurts," "I love you"—in under a second. The vendor advertises an accuracy. The number is large. And the curious reader is left with a question that turns out to be surprisingly hard to answer: *how far are we, actually, from translating what a dog says?*

This survey takes that question seriously and refuses to answer it with a single number. "Translation" is not one capability but a stack of them, and the distance to each layer is different. A system that reports whether a dog is relaxed or distressed is doing something real and useful; a system that claims to render the dog's "words" as an English sentence is doing something else, and the difference between the two is not a matter of degree but of kind. Our central move is to make that stack explicit—a five-rung *translation ladder* (Section 3)—and to locate current research and current products on it. The headline finding is easy to state and, we will argue, well supported: **today's best systems live on the second rung, affect classification, with a handful of careful laboratory studies reaching toward the third.** Everything above that remains either aspirational or, in the strong sense, ill-posed.

1.1 Why now: three converging currents

Three developments make this the right moment for such a survey. First, *bioacoustic foundation models* have matured. Self-supervised encoders trained on animal sound—AVES [24], the audio–language model NatureLM-audio [25], and Google DeepMind's Perch line [26]—now transfer across species and tasks, and standardized benchmarks such as BEANS [27] and BirdSet give the field common measuring sticks. Second, *flagship decoding projects* have produced concrete, peer-reviewed structure discoveries: Project CETI's sperm-whale "phonetic alphabet" [19], Google's DolphinGemma sequence model [21], and the Earth Species Project's interactive songbird work. Third, *canine cognition* has its own empirical renaissance, from the ELTE Budapest group's decades of bark and growl psychophysics [1, 4] to the UC San Diego soundboard studies that put "talking buttons" under controlled observation [15, 16]. The intersection of these three currents is exactly the space this paper maps.

1.2 The thesis in one paragraph

A dog's communicative repertoire is rich, multimodal, and genuinely informative—but it is, in the technical sense, a *signaling system* rather than a *language*. It encodes affect, arousal, motivational state, individual identity, and some referential content (a specific toy, a specific door), yet it lacks the open-ended, recursive, generative syntax that human language translation presupposes. Consequently, "translating what a dog says" is, above the level of affect and intent, partly a *category error*: there may be no sentence-shaped meaning in the signal to recover. This is not a counsel of despair. It reframes the achievable goal—validated, calibrated affect-and-intent inference—and exposes why the field's most quotable numbers ("81.5% accuracy," "94.6% accuracy") are, as presently defined, close to meaningless: there is no ground-truth intended message to score a translation against. We call this the *falsifiability problem*, and it is the

methodological heart of the survey (Section 7).

1.3 Contributions and organization

This paper makes four contributions.

(1) **A conceptual ladder** that separates signal, affect, intent, reference, and dialogue, and uses it to adjudicate over-claiming (Section 3, Figure 1).

(2) **A unified technical synthesis** of the decoding pipeline—encoding, structure discovery, grounding, validation—connecting canine work to the broader cross-species literature (Sections 2–5).

(3) **A critical audit** of commercial "pet translators" and the evaluation crisis that surrounds them (Sections 6–7).

(4) **An honest distance estimate and roadmap**, with confidence bands and an ethics framework (Sections 8–10, Figure 2).

Section 2 grounds the discussion in what dogs actually signal; Section 11 concludes. Throughout, we keep two registers separate: what is *measured* (peer-reviewed effect sizes, model metrics on defined tasks) and what is *claimed* (vendor accuracies, press-release superlatives). The gap between those registers is, in a real sense, the subject of this paper.

2 What Dogs Actually Signal: The Biological Substrate

Before asking whether a machine can translate a dog, we must ask what there is to translate. Canine communication is multimodal, graded, and heavily context-dependent. It is also, crucially, *asymmetric*: much of what a dog "says" is addressed to humans, and the dog's receptive understanding of human signals far exceeds its productive repertoire. This section reviews the channels and the evidence that each carries decodable information. Table 1 summarizes the landscape.

2.1 Vocalizations: barks, growls, whines

The domestic dog is unusually vocal for a canid, a trait widely interpreted as a product of domestication and paedomorphosis. The ELTE Budapest ethology group has built the most systematic body of evidence on what barks encode. Pongrácz and colleagues showed that human listeners can classify barks recorded in different behavioral situations above chance [1], and that acoustic parameters—fundamental frequency, tonality, and inter-bark interval—carry emotional information that listeners reliably extract [2]. Later work established that barks also carry *individual identity*: listeners can discriminate dogs from bark acoustics alone [3], and machine classifiers applied to thousands of barks from a kennel of Hungarian sheepdogs recovered both context and individual at rates well above chance, with "stranger at the fence" and "fight" contexts most separable and "play" least. Growls are even more informative: Faragó et al. demonstrated that human listeners judge a dog's size and aggressive intent from growl formants, and that growls produced in food-guarding versus play contexts are acoustically and perceptually distinct [4]. Whines and whimpers, associated with separation and submission, complete a repertoire that is graded rather than discrete.

The engineering literature has begun to automate exactly these judgments. The ICASSP 2018 study "What is my dog trying to tell me?" framed bark analysis as automatic recognition

of *context* and *perceived emotion* [5], explicitly building on the ELTE psychophysics. The important nuance—which we will return to in Section 7—is that the target label is the *human-perceived* emotion or the *eliciting context*, not the dog's "intended message." The model learns a mapping that humans already agree on; it does not recover a hidden sentence.

2.2 The face: DogFACS and the "puppy dog eyes" effect

Canine facial behavior is codable with the same rigor as human facial action coding. DogFACS, the Dog Facial Action Coding System [6], adapts Ekman and Friesen's FACS to the musculature of the dog face, defining action units (e.g., AU101, the inner brow raiser) and ear action descriptors. Using DogFACS, Kaminski et al. showed that dogs produce markedly more facial movement—especially the inner brow raiser that enlarges the apparent eye—when a human is *attending* to them than when the human's back is turned, and that the presence of food does not drive the effect [7]. This is a striking result: it suggests the "puppy dog eyes" expression is a *communicative* signal tuned to human attention, possibly shaped by selection during domestication, rather than a mere reflex of internal state. A 2024 comparative study further reported that domestication *constrains* the dog's facial expressiveness relative to wolves [8], complicating any simple "dogs emote freely" narrative.

Automated facial analysis is now an active computer-vision subfield. Explainable CNN pipelines recognize emotional states from canine facial action patterns [9], improved optimization schemes push expression-recognition accuracy [10], and 2025 work automates emotional-expression analysis end to end [11] and standardizes facial landmark detection [12]. These are genuine advances—but, again, their labels are human-coded affective categories or DogFACS action units, not propositions.

2.3 The body: posture, tail, and gaze

A large fraction of everyday canine signaling is postural: ear position, tail carriage and wag direction, weight distribution, piloerection, and gaze. Classic work reported lateralized tail-wagging biases tied to approach versus withdrawal motivation, and the broader ethogram of dominance, submission, play (the play bow), and stress is well documented in the clinical literature. Body signals are often *more* diagnostic than vocalizations, a point commercial vendors themselves concede: Traini's founder has stated that audio alone is insufficient and that combining body language and facial expression raises accuracy. This is an implicit admission that the "translation" is really multimodal affect classification. Olfactory signaling—urine marking, pheromonal communication—is enormously important to dogs but essentially inaccessible to current consumer sensors, and we flag it as a blind spot rather than a solved channel.

2.4 Reception: dogs understand more than they say

Any honest account must separate what dogs *produce* from what they *comprehend*. The receptive side is impressive. Rico, a border collie, provided the first evidence of "fast mapping"—inferring a novel word's referent by exclusion—in *Science* [13]. Chaser, another border collie, was documented retrieving 1,022 named objects and responding to simple noun–verb combinations [14]. These are exceptional, intensively trained individuals, but they establish an upper bound on referential comprehension that is far higher than the productive repertoire. The

Table 1: Canine communication channels, the information they carry, and the strongest evidence. "Decodable today" refers to validated, above-chance machine or human inference on a defined label—not to sentence-level translation.

Channel	Information encoded	Key evidence	Decodable today
Bark	Affect, arousal, context, individual identity	Pongrácz et al. [1, 2]; Molnár et al. [3]	Affect/context (above chance)
Growl	Body size, aggressive intent, resource-guarding vs. play	Faragó et al. [4]	Size/intent (human + ML)
Whine/whimper	Separation distress, submission	Ethological observation	Coarse affect
Face (DogFACS)	Attention-directed signals, affective state	Waller et al. [6]; Kaminski et al. [7]	Action units; affect (CNN)
Body/tail/gaze	Motivation, stress, play, social intent	Clinical ethograms; wag-laterality studies	Posture classes
Olfaction	Identity, reproductive state, territory	Scent-marking literature	*Not* sensor-accessible
Reception (input)	Human word referents (up to 1,022)	Kaminski et al. [13]; Pilley & Reid [14]	Comprehension, not production
Soundboard (AIC)	Referential intent via human symbols	Bastos et al. [15, 16]	Word-level intent (trained)

asymmetry matters: a future "dialogue" system may succeed by letting dogs *receive* structured human input (as soundboards do) long before it can *read* structured canine output.

2.5 Soundboards: the closest thing to canine "speech"

The most direct attempt to give dogs a productive symbolic channel is the augmentative interspecies communication (AIC) soundboard: a grid of buttons that each speak a recorded word when pressed. Popularized by speech pathologist Christina Hunger and the Instagram dog Bunny, the phenomenon is now under controlled study by Federico Rossano's UC San Diego *TheyCanTalk* project, which tracks thousands of button-using pets across dozens of countries. Two 2024 peer-reviewed papers anchor the science. Bastos et al. (*PLoS ONE*) showed that soundboard-trained dogs respond appropriately to words such as "play" and "outside" whether the word is spoken by the owner or triggered by an unfamiliar button-presser, indicating word comprehension rather than cue-reading [15]. A companion *Scientific Reports* paper analyzed two-button combinations and found them to be non-accidental, non-random, and non-imitative—evidence of intentional, if linear, combination [16]. A 2025 review synthesizes the evidence and the skepticism [17]. The consensus interpretation is careful: the combinations look like associative sequences (name + action, place + go), not recursive grammar; spatial and immediate concepts ("outside," "paw") are robust, while temporal abstractions ("yesterday," "tomorrow") are rare and inconsistent. Soundboards thus demonstrate *referential intent* on the third rung of our ladder—but they do so by giving the dog a human-designed symbol set, not by decoding the dog's natural signal.

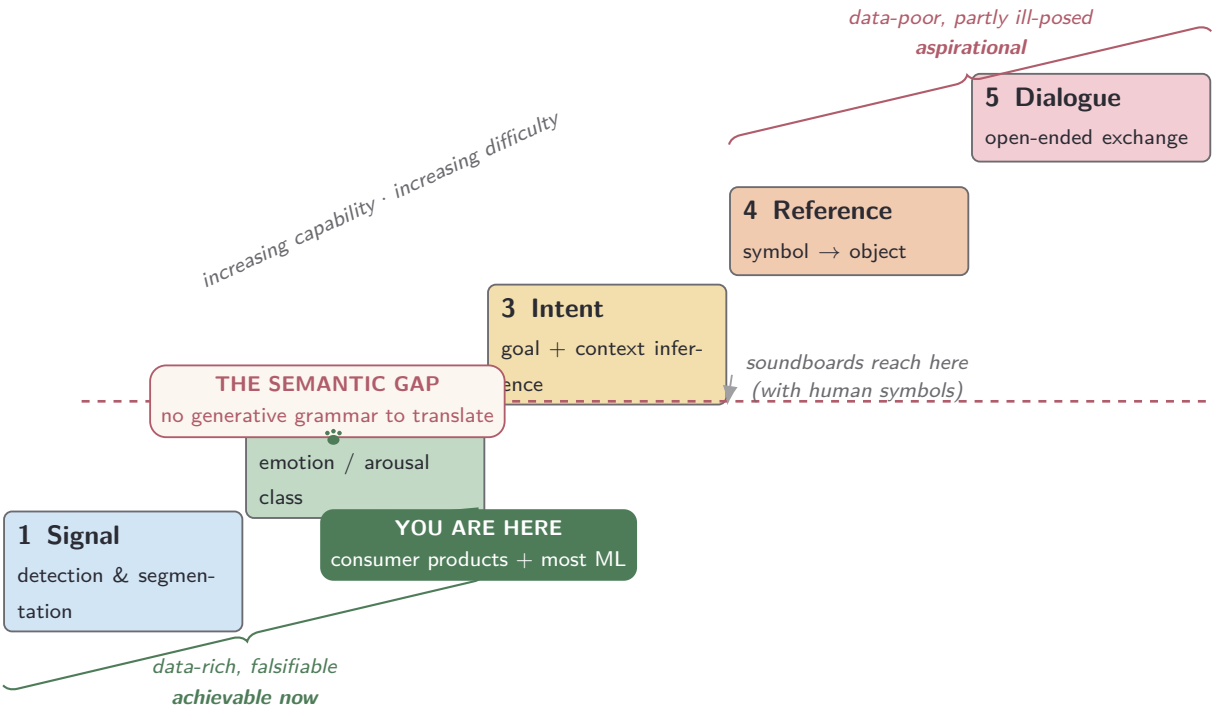


Figure 1: The translation ladder. "Translating a dog" decomposes into five capabilities of increasing difficulty. Current consumer products and most machine-learning work occupy **Rung 2** (affect classification); controlled soundboard studies reach **Rung 3** (referential intent) but only by supplying a human-designed symbol set. The dashed boundary marks the *semantic gap*: above it, the dog's natural signal lacks the recursive, generative syntax that sentence-level translation presupposes, so the target becomes partly ill-posed rather than merely hard. The lower rungs are data-rich and falsifiable; the upper rungs are data-poor and, in the strong sense, may have no ground truth to recover.

2.6 The takeaway

The biological substrate is real and rich: dogs signal affect, arousal, identity, motivation, and—with human scaffolding—referential intent. What the substrate does *not* contain is an open-ended generative grammar. This single fact, more than any limitation of current models, bounds what "translation" can mean. We formalize that bound next.

3 The Translation Ladder: What "Translating a Dog" Could Mean

The phrase "translate what a dog says" smuggles in an assumption: that there is a sentence-shaped meaning waiting in the signal, the way a French sentence waits behind its English translation. The biological review of Section 2 gives us reason to doubt this. To make progress we replace the single verb "translate" with a ladder of five distinct capabilities, each with its own input–output contract, its own notion of ground truth, and its own distance from the present (Figure 1).

3.1 Rung 1: signal detection and segmentation

The base rung asks only: *is there a vocalization, where does it begin and end, and who produced it?* This is a classical detection-and-classification problem with objective ground truth (a human annotator marks the waveform), and it is essentially solved. Bioacoustic detectors trained on large corpora [26, 24] segment and attribute calls reliably; individual identification from bark acoustics is above chance [3]. Nothing about Rung 1 requires understanding.

3.2 Rung 2: affect and arousal classification

The second rung maps a signal to a small set of affective categories—happy, fearful, aggressive, distressed, relaxed—or to continuous arousal/valence dimensions. This is where nearly all commercial "translators" actually live, and where the peer-reviewed ML literature is strongest, because the labels are human-perceived emotions or eliciting contexts that annotators can agree on [2, 5, 9]. Rung 2 is *falsifiable*: one can hold out recordings, score against consensus labels, and report a real accuracy. It is also genuinely useful for welfare monitoring. Its limit is that an emotion label is not a sentence; rendering "fearful" as "I am scared of the vacuum" adds narrative the signal never contained.

3.3 Rung 3: intent and context inference

The third rung asks what the dog is *trying to do*—solicit play, request egress, guard a resource, signal pain—given the surrounding context. This requires multimodal fusion (voice + face + posture + situation) and begins to need a model of the dog's goals. Soundboard studies provide the cleanest evidence here, because the button press makes the intent explicit and checkable [15, 16]. But note the scaffolding: the symbol set is human-designed and trained. Rung 3 on *natural* signals is far less developed and much harder to validate.

3.4 Rung 4: referential semantics

The fourth rung would map a spontaneous canine signal to a specific referent—*that* ball, *this* door, a particular person—without human-supplied symbols. Receptive studies show dogs can *understand* such mappings [13, 14], and some vocal signals carry identity information, but a productive, open referential vocabulary in natural dog communication has not been demonstrated. This is the first rung where the "translation" framing starts to strain: there may be no stable symbol-to-referent dictionary to recover.

3.5 Rung 5: open-ended dialogue

The top rung is the folkloric dream: free, generative, two-way conversation in which the dog expresses novel propositions and understands ours. It presupposes a recursive generative grammar on the canine side. The evidence of Section 2—graded affective signals, linear associative combinations, no syntax—indicates this presupposition is false for natural canine communication. Rung 5 is therefore not merely distant; in its strong form it is *ill-posed*. A weaker, engineered version (structured human-to-dog input via soundboards, with the dog responding through trained symbols) is conceivable and already nascent.

3.6 Why the ladder matters

The ladder converts a vague question into answerable ones. "Can we translate a dog?" becomes: *which rung, with what ground truth, at what validated accuracy?* It also explains the rhetoric of the market: vendors demonstrate Rung 2 while marketing Rung 5. The rest of this survey uses the ladder as a ruler—to measure the technical pipeline (Section 4), to read the cross-species evidence (Section 5), and to audit the products (Section 6).

4 The Technical Pipeline: From Waveform to (Something Like) Meaning

If genuine translation were possible, what would the system look like? Drawing on the cross-species decoding literature, we decompose the problem into four stages—*encoding*, *structure discovery*, *semantic grounding*, and *interactive validation*—and assess the maturity of each for the canine case. The stages are not a waterfall so much as a loop: validation feeds back into grounding, which constrains structure, which shapes the encoder.

4.1 Stage 1: self-supervised acoustic encoding

The modern starting point is a self-supervised representation trained on large, unlabeled audio. AVES (Animal Vocalization Encoder based on Self-Supervision) adapts the HuBERT/wav2vec paradigm to animal sound and shows that representations learned without labels transfer to downstream bioacoustic tasks [24]. NatureLM-audio extends this to an *audio–language* foundation model: trained on text–audio pairs spanning bioacoustics, speech, and music, it can detect vocalizations, classify rare species, and even caption or generate animal sounds from natural-language prompts, demonstrating transfer from speech and music to bioacoustics [25]. Google DeepMind's Perch line and the global birdsong embeddings show that a single encoder can serve many taxa [26]. Standardized benchmarks—BEANS (the Benchmark of Animal Sounds) and BirdSet—give these encoders a common evaluation surface [27], while open general-purpose encoders such as OpenBEATs push state of the art on a majority of BEANS subtasks [28]. For dogs, the implication is that a strong *Rung 1–2* encoder is available off the shelf; the bottleneck is not representation quality but labels and semantics.

4.2 Stage 2: sequence-structure discovery

Given an encoder, the next question is whether the signal has *combinatorial structure*—discrete units that recombine, the way phonemes form words. This is where the flagship projects have produced their most cited results. Project CETI's analysis of 8,719 sperm-whale codas revealed a "phonetic alphabet": rhythm, tempo, rubato, and ornamentation combine contextually, expanding the repertoire of distinguishable forms by nearly an order of magnitude [19]. Crucially, the authors are explicit that this describes *form*, not *meaning*: the paper answers what the units are and how they combine, not what most of them say. Google's DolphinGemma takes the language-model route directly—a roughly 400-million-parameter Gemma-based model with a SoundStream tokenizer, trained on the Wild Dolphin Project's four-decade Atlantic spotted dolphin corpus, that predicts the next sound in a sequence exactly as an LLM predicts the next token [21]. CETI's WhAM (2025) similarly learns to complete masked clicks and to render

whale-like codas from other audio. The Earth Species Project's 2026 zebra-finch model (ZF-AIM) goes further, detecting, predicting, and *generating* calls in real time and engaging birds in turn-taking exchanges.

The lesson for dogs is double-edged. On one hand, the methods are species-agnostic and could be applied to bark sequences. On the other, the structure these models find is *statistical*, and predicting the next sound is not understanding it. Yossi Yovel and others have warned that such models may cluster calls by acoustic similarity while missing meaning entirely—grouping signals that sound alike but signify different things. For a repertoire as graded and context-bound as the dog's, the prior probability of finding a clean combinatorial alphabet is lower than for whale codas or birdsong.

4.3 Stage 3: semantic grounding

Structure without semantics is syntax without a dictionary. Grounding ties a signal to the world: to the eliciting context, the dog's subsequent behavior, the objects in view, the social partner. This is the hardest stage and the one where dogs are paradoxically *advantaged*, because they live in richly instrumented human homes and their behavior is legible to us. The soundboard paradigm is effectively a grounding device [15, 16]: the button press, the spoken word, and the consequent action (door opens, toy appears) form a triangulated, checkable mapping. The Wild Dolphin Project's CHAT system does the analogous thing underwater, pairing a synthetic whistle with a coveted object (sargassum, a scarf) so that the dolphin can request it by name [21]. The general principle—*ground symbols through interaction and consequence*, not through passive correlation—is the field's most important methodological insight, and it is exactly what consumer bark-to-sentence apps lack.

4.4 Stage 4: interactive validation (playback)

The final stage closes the loop: does the animal treat the model's output as meaningful? The gold standard is the *playback experiment*—play a recorded or synthesized signal and observe whether the receiver responds appropriately. Comparative ethology has used playbacks for decades to establish that, e.g., vervet alarm calls refer to specific predators. The Earth Species Project's interactive finch work and the CHAT dolphin protocol are modern, model-driven versions. For dogs, a validation loop would look like: infer an intent, emit a candidate signal or action, and test whether the dog's behavior confirms the inference—with the experimenter blind to the model's guess to avoid Clever-Hans contamination. No consumer product does this; it is the dividing line between science and entertainment.

4.5 Maturity summary

Table 2 rates each stage for the canine case. The pattern is clear: the *front* of the pipeline (encoding, structure) is borrowable from the broader bioacoustics boom and is relatively mature; the *back* (grounding, validation) is where canine-specific, falsifiable work is needed and where almost none of the commercial effort is directed. This asymmetry—strong models, weak semantics, absent validation—is the technical signature of the whole field.

Table 2: Maturity of the four decoding stages for the canine case. "Borrowable" means general bioacoustic methods transfer directly; "canine-specific" means dedicated dog data and protocols are required.

Stage	Representative method	Maturity	Canine status
Encoding	AVES, NatureLM-audio, Perch [24, 25, 26]	High	Borrowable; Rung 1–2 ready
Structure	CETI alphabet, DolphinGemma, WhAM [19, 21]	Medium	Statistical only; meaning open
Grounding	Soundboards, CHAT object-labeling [15, 21]	Low–medium	Needs interaction data
Validation	Playback / closed-loop response	Low	Absent in products

5 Lessons from Other Species: Whales, Dolphins, and Finches

Dogs are not the first species for which decoding has been attempted, and the comparison is instructive. The flagship projects differ from the canine case in one decisive respect: their subjects often possess *discrete, culturally transmitted vocal units* (whale codas, dolphin signature whistles, birdsong syllables) that invite combinatorial analysis. Dogs, by contrast, signal with graded, context-bound affective displays. Reading the cross-species literature through the ladder of Section 3 clarifies both what is transferable and what is not. Figure 2 places these milestones on a capability timeline; Table 3 gives the details.

5.1 Project CETI: structure is not meaning

The Cetacean Translation Initiative is the field's most ambitious program. Its 2024 *Nature Communications* analysis of 8,719 codas from roughly 60 East Caribbean sperm whales demonstrated a context-dependent combinatorial system—rhythm, tempo, rubato, and ornamentation recombine to expand the form inventory by nearly an order of magnitude [19]. The 2022 program paper lays out the interdisciplinary ambition [20], and the 2025 WhAM model learns to complete and generate whale-like codas [22]; a late-2025 preprint even proposes a translative model of whale vocalization [23]. Yet the investigators are careful: the "phonetic alphabet" describes how signals are *built*, not what they *mean*. For dogs, the transferable lesson is the methodology (large corpora, structure discovery, playback validation); the *non*-transferable part is the assumption of a discrete combinatorial inventory, which canine graded displays may simply lack.

5.2 DolphinGemma and CHAT: grounding through interaction

Google DeepMind's DolphinGemma, with Georgia Tech and the Wild Dolphin Project, is a roughly 400-million-parameter audio sequence model built on Gemma with a SoundStream tokenizer, trained on four decades of Atlantic spotted dolphin recordings and small enough to run on a Pixel phone [21]. Its companion CHAT device does not try to translate natural whistles directly; instead it assigns *synthetic* whistles to coveted objects and lets dolphins request them by name—a grounding-through-interaction protocol. This is the single most important design lesson for canine work: meaning is established by *consequence* (the dolphin gets the sargassum), not by passive correlation. A dog soundboard is the terrestrial analogue.

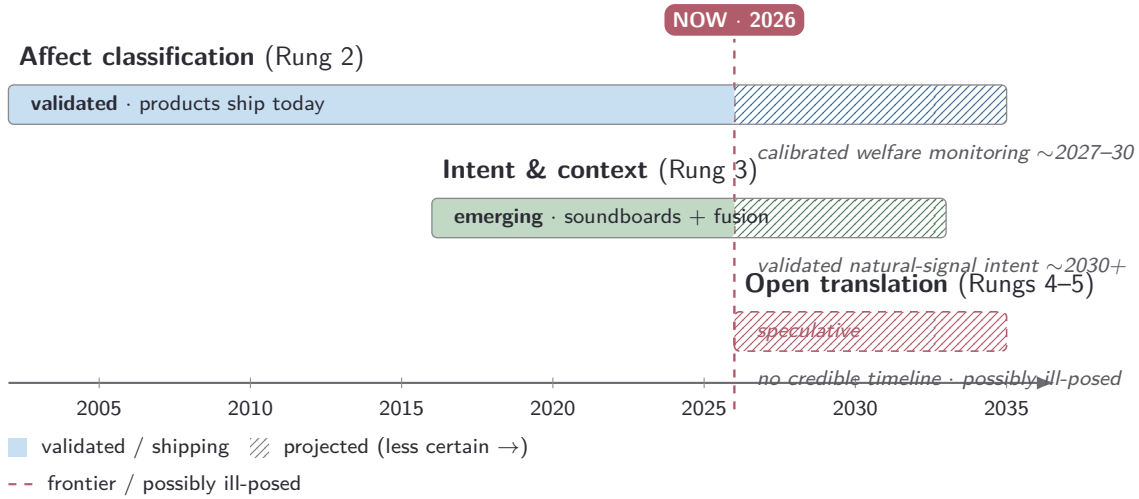


Figure 2: Capability roadmap, 2002–2035. Three tracks track the translation ladder over time. Solid bars denote validated, shipping capability; hatched bars denote projection, with confidence decreasing to the right. **Track A** (affect classification, Rung 2) is mature today and heads toward calibrated welfare monitoring. **Track B** (intent and context, Rung 3) is emerging through soundboards and multimodal fusion, with validated natural-signal intent plausible after 2030. **Track C** (open referential translation and dialogue, Rungs 4–5) has no credible timeline and may be permanently ill-posed for natural canine communication. The vertical dashed line marks the present (2026). Method-enabling milestones (CETI, DolphinGemma, foundation models) are detailed in Table 3; they advance the *tools*, not the *semantics*.

5.3 The Earth Species Project: from encoding to interaction

The Earth Species Project (ESP), founded in 2018, supplies much of the field's open infrastructure: the AVES self-supervised encoder [24], contributions to NatureLM-audio and the BEANS benchmark [25, 27], and a 2026 zebra-finch model (ZF-AIM) that detects, predicts, and generates calls in real time, engaging birds in turn-taking exchanges [29]. ESP's trajectory—encode, then discover structure, then *interact*—is the pipeline of Section 4 made flesh. It also surfaces the standing critique: Yossi Yovel and others caution that models may over-rely on acoustic similarity, clustering calls that sound alike but mean different things [30].

5.4 What transfers to dogs—and what does not

Three things transfer: the self-supervised encoders, the structure-discovery toolkit, and above all the interaction-grounding/validation protocol. Three things do not: the assumption of discrete culturally transmitted units, the luxury of decades of labeled wild corpora, and the clean playback accessibility of whales and birds. Dogs are domestic, multimodal, and graded; their advantage is legibility and proximity, their disadvantage is the absence of a syntax to decode. The cross-species evidence thus *supports* the ladder's verdict: the tools for Rungs 1–3 are arriving fast, while Rungs 4–5 await a discovery about canine semantics that may never come, because there may be nothing of that kind to discover.

Table 3: Selected milestones in cross-species communication decoding. "Type" distinguishes peer-reviewed structure results (R), generative / foundation models (M), interactive grounding protocols (I), and consumer products (P). These advance the toolbox; none yet delivers validated sentence-level translation for any species.

Year	Milestone	Type	Significance
2002	Bow-Lingual [32]	P	Six-emotion bark matcher; Ig Nobel; the consumer archetype
2005	ELTE bark psychophysics [1]	R	Humans classify bark context above chance
2013	DogFACS [6]	R	Objective facial-action coding for dogs
2018	ESP founded; ICASSP bark ML [5]	M/R	Open infrastructure; automated context/affect
2022	CETI program paper [20]	R	Roadmap for whale decoding
2023	AVES; BEANS [24, 27]	M	Self-supervised animal-sound encoder + benchmark
2024	Whale phonetic alphabet [19]	R	Combinatorial coda structure (form, not meaning)
2024	Soundboard studies [15, 16]	R/I	Canine word comprehension + intentional combination
2024	NatureLM-audio [25]	M	Audio–language foundation model for bioacoustics
2025	DolphinGemma + CHAT [21]	M/I	Sequence model + object-grounded interaction
2025	WhAM [22]	M	Generative whale-coda model
2026	ZF-AIM finch interaction [29]	I	Real-time turn-taking with a songbird

6 The Commercial Landscape: Translators That Are Not Translators

A small industry now sells the folkloric dream as a feature. This section catalogs the principal products, reports their claims *as claims*, and audits what they actually do against the ladder. The pattern is uniform: the technology is affect classification (Rung 2); the marketing is dialogue (Rung 5); and the accuracy numbers, as defined, cannot be checked. Table 4 summarizes the landscape.

6.1 A brief history: from Bow-Lingual to PetGPT

The archetype is Takara's Bow-Lingual (2002): a collar microphone and handheld display that matched bark acoustics to six emotions—happiness, sadness, frustration, anger, assertion, desire—and then to one of roughly 200 canned phrases [32]. It sold well, won the 2002 Ig Nobel Peace Prize, and established the template: *classify affect, then decorate it with a sentence.* Two decades later the template is unchanged; only the classifier has been swapped for a deep model and the phrase bank for a language model.

The current flagship is Traini, a Silicon Valley "pet empathy AI" whose PetGPT/PEBI model analyzes sound, images, and video to infer canine emotion and need [31]. Vendor materials report recognition of about 12 emotions across roughly 120 breeds, with a headline accuracy

of 81.5%, and describe a corpus on the order of hundreds of thousands of dogs' sounds and expressions labeled with input from behavior specialists. The company has raised venture funding and amassed a large consumer user base. Notably, its founder has publicly conceded that audio alone is insufficient and that body language and facial expression must be fused to reach the claimed accuracy—an implicit admission that the product is multimodal affect inference. The reverse direction (human speech rendered as "dog") is reported to be far weaker, on the order of 40% in independent coverage, and rests on the questionable premise that synthesized barks carry meaning to the dog.

6.2 The wider field

Other products fill out the picture. MeowTalk, from a former Amazon engineer, applies the same idea to cats [33]. The Petpuls collar, a Seoul National University spin-off, classifies emotional states from vocalizations [35]. In China, "Meng Xiao Yi" reports a Qwen-based model trained on millions of pet voiceprints with a claimed 94.6% emotion accuracy [34], and vendors such as PETKIT and various "pet LLM" startups market health-and-emotion monitoring. DogTok and similar apps invert the direction, turning human commands into bark-like sounds for training. The market context is a global pet industry measured in hundreds of billions of dollars, with smart devices the fastest-growing segment—which explains the velocity of the claims, if not their validity.

6.3 The audit: what the accuracy numbers actually measure

Read carefully, every headline number is an *emotion-classification* metric, not a translation metric. "81.5%" means: on some held-out set, the model's emotion label agreed with a human-assigned label 81.5% of the time. That is a real, falsifiable Rung-2 result—*if* the test set, label protocol, and class balance are disclosed, which consumer vendors rarely do. It says nothing about whether the rendered sentence ("I love you forever") corresponds to anything the dog intended, because no such ground truth exists. The sentence is generated *downstream* of the classifier by a language model optimizing for human emotional resonance, not for fidelity to the animal. This is the commercial instance of the semantic gap: the product demonstrates Rung 2 and sells Rung 5.

6.4 Why the gap persists

Three incentives keep the gap open. First, the consumer value proposition is emotional—owners want to feel understood by their pets—so a plausible, affectionate sentence beats a calibrated probability vector, even though the latter is the honest output. Second, the data flywheel favors engagement over validation: every uploaded clip enriches the classifier, but no mechanism tests whether the "translation" is correct, because the dog cannot complain. Third, the scientific validation loop (playback, blind protocols) is slow and unglamorous, and yields no shareable screenshot. The result is a market optimized for the *appearance* of translation. Section 7 formalizes why this appearance cannot, as presently constructed, be falsified.

Table 4: Commercial "pet translators" audited against the ladder. Accuracies are *vendor-reported claims*, not independently verified; all measure emotion classification (Rung 2), none measures sentence-level translation. "Reverse" = human-to-animal direction.

Product	Modality / method	Claim	Actual rung	Reverse
Bow-Lingual (2002) [32]	Bark acoustics → 6 emotions	∼200 phrases	2	No
Traini / PetGPT [31]	Audio + image + video, PEBI	81.5%, 12 emotions	2	Weak (∼40%)
MeowTalk [33]	Cat vocalization classifier	Emotion labels	2	No
Petpuls [35]	Collar vocalization analysis	Emotion states	2	No
Meng Xiao Yi [34]	Qwen + voiceprints	94.6% (claimed)	2	No
DogTok	Human command → bark	Training aid	n/a	Yes

7 The Evaluation Crisis: Why the Numbers Cannot Be Trusted

If the ladder tells us *what* could be translated, the evaluation problem tells us why we cannot yet know *whether* it has been. This section argues that the field's most quoted quantities—vendor accuracies above 80%, press-release superlatives—are, as presently defined, *unfalsifiable*, and that fixing this is a prerequisite for calling any system a translator.

7.1 The ground-truth problem

Translation, in the human sense, is scored against a reference: a bilingual corpus supplies the intended meaning, and metrics such as BLEU or human adequacy judgments measure fidelity to it. Canine communication has no such reference. There is no canonical "intended message" behind a bark, no bilingual informant who can confirm that "I want to play" is what the dog meant. The dog cannot report its intent, and—per Section 2—there may be no sentence-shaped intent to report. Consequently, a bark-to-sentence system cannot be scored for translational fidelity at all. What *can* be scored is the upstream classifier: does the model's emotion label agree with a human consensus label on a held-out recording? That is a Rung-2 metric, and it is the only honest number in the pipeline. Reporting it as a "translation accuracy" is a category mistake.

7.2 What the vendor numbers actually are

When a product advertises "81.5% accuracy," the defensible reading is: on some internal test set, the predicted emotion class matched a human-assigned class 81.5% of the time. Three problems follow. First, the test set, label protocol, class balance, and inter-annotator agreement are rarely disclosed, so even the Rung-2 claim cannot be independently checked. Second, the label is the *human-perceived* emotion or the *eliciting context*, which is a proxy for the dog's state, not a readout of it; the ELTE tradition is explicit about this [2, 5]. Third, the rendered sentence is produced *after* classification by a language model optimized for human resonance, so its fidelity

is never measured at all. The number thus travels down the pipeline as marketing while its meaning stays fixed at the top.

7.3 Unfalsifiability and the absent complaint

A scientific claim must be falsifiable: there must be some observation that could show it false. "This bark means 'I love you' " admits no such observation—the dog will not contradict it, and any subsequent behavior can be retrofitted to the narrative. This is the structural reason the consumer market cannot self-correct: there is no error signal. Compare a weather forecast, which is scored against tomorrow's weather, or a speech-to-text system, scored against a transcript. The pet translator has no tomorrow and no transcript. Until a validation loop supplies an error signal, its outputs are unfalsifiable interpretations, not measurements.

7.4 Clever Hans, anthropomorphism, and the observer effect

Two classic threats compound the problem. The *Clever Hans* effect—the horse that appeared to do arithmetic but was reading its trainer's unconscious cues—is the permanent hazard of any animal-communication experiment: the human observer leaks the answer. Soundboard studies guard against this with blind protocols and unfamiliar button-pressers [15]; consumer apps, used by the devoted owner in an uncontrolled home, do the opposite. *Anthropomorphism* is the second threat: attributing human propositional attitudes ("she says she loves me") to graded affective signals. The critical-review literature on canine emotion perception warns that human observers project their own categories onto the dog [9], and a language model trained on human text will amplify exactly this projection. The product thus risks teaching owners to *ignore* the dog's real body signals in favor of a flattering sentence—a welfare harm we return to in Section 9.

7.5 What a valid evaluation would require

A defensible evaluation program has four ingredients.

(1) **Defined labels and disclosed protocols.** Report Rung-2 metrics on a public, held-out corpus with inter-annotator agreement, class balance, and per-class confusion—the BEANS/BirdSet discipline applied to dogs [27].

(2) **Calibration, not just accuracy.** Output probabilities that match observed frequencies, so a welfare tool can threshold responsibly.

(3) **Closed-loop validation.** Test inferred intent against the dog's subsequent behavior under blind conditions—the playback logic of Section 4, adapted to the home.

(4) **Separate the sentence from the score.** Never report a generative rendering as evidence of fidelity; treat it as a user-interface layer, clearly labeled as interpretation.

None of today's consumer products satisfies these. The research community satisfies (1) and (2) in patches and (3) in the soundboard and cross-species literature. The gap between the two is the evaluation crisis.

7.6 A falsifiability criterion for the field

We propose a simple demarcation: *a canine communication system should be called a translator only if its outputs are scored against an independent, animal-grounded error signal.* By this

criterion, no current product qualifies; the soundboard paradigm qualifies at the word level; and the rest is affect classification with a narrative interface. This is not pedantry. It is the difference between a measurement and a story, and the field's credibility—and the animals' welfare—depends on keeping them apart.

8 So How Far Are We? An Honest Distance Estimate

We can now answer the title question without evasion. The answer is not a number of years but a map: the distance depends entirely on which rung of the ladder one means, and the rungs differ not by degree but by kind. Table 5 gives our estimate, with the evidence and the dominant blocker for each.

8.1 Rung 1–2: essentially here, and useful

Signal detection and affect classification are solved in the laboratory and shipping in products. The remaining work is engineering and validation—public benchmarks, calibration, multimodal fusion—not discovery. A calibrated, validated welfare monitor that reports arousal and valence with honest uncertainty is plausibly a **2–4 year** horizon from the research frontier to reliable deployment. This is the genuinely useful payoff: early stress or pain detection, shelter welfare assessment, veterinary triage. It is not "translation," and calling it such would oversell it; but it is real.

8.2 Rung 3: a foothold, a decade of work

Intent and context inference on *natural* signals is emerging. The soundboard literature proves that referential intent exists and is checkable [15, 16], but it does so with human-supplied symbols. Extending this to spontaneous barks, faces, and postures—fusing modalities, modeling goals, validating closed-loop—is a research program, not a product cycle. We estimate **validated natural-signal intent inference at roughly 2030 or later**, contingent on the evaluation infrastructure of Section 7 being built. The blocker is not model capacity; it is labeled, grounded, validated data.

8.3 Rung 4–5: possibly never, and that is informative

Open referential translation and free dialogue presuppose a generative symbolic system in the dog that the biological evidence does not support. Here the honest answer is **no credible timeline**, and the reason is not pessimism about AI but a fact about the subject: there may be no language there to translate. This is the survey's central, deflationary claim. It does not rule out an *engineered* dialogue—structured human-to-dog input via trained symbols, with the dog responding through a soundboard—which is a Rung-3 artifact wearing Rung-5 clothing. But faithful translation of spontaneous canine "speech" into English sentences is, in the strong sense, ill-posed, and more compute will not fix a category error.

8.4 Why "more scale" is not the answer

It is tempting to read the cross-species successes as evidence that scaling foundation models will eventually crack the dog. The reading is mistaken. CETI, DolphinGemma, and the finch models advance *structure discovery and generation*—the front of the pipeline. They do not

Table 5: Distance estimate by rung. Horizons are judgments from the evidence reviewed, not predictions; "blocker" names the dominant obstacle. Rungs 4–5 are marked ill-posed because the requisite generative symbolic system is not evidenced in natural canine communication.

Rung	Status	Horizon	Dominant blocker
1 Signal	Solved	Now	Engineering only
2 Affect	Shipping	2–4 yr to validated welfare use	Calibration, public benchmarks
3 Intent	Foothold	∼2030+	Grounded, validated data
4 Reference	Unproven	No credible timeline	No productive symbol system
5 Dialogue	Ill-posed	Possibly never	No generative grammar in subject

supply semantics, and their authors say so [19, 21]. The canine bottleneck is at the *back* of the pipeline (grounding, validation) and, above Rung 3, in the subject itself. Scaling the encoder makes Rung 1–2 better; it does not create a grammar that is not there. The distance to Rung 5 is therefore not a function of parameters but of biology.

8.5 The reframe: from translation to interpretation

The productive move is to stop asking "when will we translate dogs?" and start asking "how well can we *interpret* canine state, with calibrated uncertainty and animal-grounded validation?" That question has answers, has metrics, and has welfare value. It also respects the animal: it treats the dog as a signaler with its own modalities, not as a defective human whose sentences we are waiting to recover. The Rosetta Stone metaphor that titles this survey is, in the end, a warning as much as a promise: the Rosetta Stone worked because the same decree existed in three scripts. For the dog, there may be no decree—only a rich, graded, affective life that we can learn to read, but not to transcribe.

9 Ethics and Governance: Responsible Interspecies AI

A technology that claims to read an animal's mind carries obligations that a ordinary classifier does not. This section outlines an ethics framework for canine communication AI, organized around welfare, honesty, privacy, and the moral status of the animal as a communication partner.

9.1 Welfare first: do no harm

The most immediate risk is that a flattering "translation" displaces attention to the dog's real signals. An owner told "she says she loves you" may overlook the flattened ears, the whale eye, the lip lick that signal stress; an owner told "my tummy hurts" may either over-react or, worse, dismiss genuine pain when the app stays silent. The critical-review literature on canine emotion perception is explicit that human observers already project their categories onto dogs [9]; a resonant language model amplifies the projection. A welfare-first design therefore (i) surfaces calibrated state estimates with uncertainty, not sentences; (ii) never suppresses or contradicts observable body signals; and (iii) routes high-stakes inferences (pain, aggression, distress) to

human or veterinary review. Data collection itself must be non-invasive: passive audio and video, no induced stress to elicit "training" barks.

9.2 Honesty: labeling interpretation as interpretation

The honesty obligation follows directly from the evaluation crisis (Section 7). A system must not present a generative rendering as a measurement. Concretely: separate the calibrated classifier output from the narrative interface; label the latter as interpretation; disclose that accuracy figures refer to emotion classification, not translation; and refrain from reverse-direction claims (human-to-dog "speech") that have no validation. The Ig Nobel history of Bow-Lingual is a reminder that the market rewards the appearance of understanding; governance should reward the substance.

9.3 Privacy: the home as sensor

Canine monitors live in the home, always-on, capturing audio and video that inevitably include humans. The data flywheel that improves the classifier also accumulates intimate domestic recordings. Privacy-by-design—on-device inference where possible (the DolphinGemma-on-Pixel precedent is instructive [21]), minimal retention, explicit consent for cloud training, and clear deletion—is not optional. The animal cannot consent; the human household must, and the design should make that consent meaningful.

9.4 The animal as partner, not oracle

Finally, there is a conceptual obligation. Treating the dog as an oracle whose "words" we extract instrumentalizes it and flattens its communicative life into a human format. The soundboard paradigm is ethically attractive precisely because it treats the dog as a *partner* in a shared, negotiated symbol system, with the dog's behavior as the check [15, 16]. The cross-species community is beginning to formalize this: CETI's leadership has stressed that any breakthrough in whale communication should be a collective human decision, with responsible-interaction guidelines developed before signals are sent to wild animals. The canine analogue is to validate through the dog's response, to avoid anthropomorphic over-claiming, and to keep the animal's welfare as the terminal criterion.

9.5 A governance checklist

We summarize the framework as a checklist for builders and reviewers:

(1) Report Rung-2 metrics on public, held-out data with calibration; never label them "translation accuracy."

(2) Separate interpretation (narrative) from measurement (classifier); label both clearly.

(3) Validate intent claims closed-loop and blind; otherwise call them hypotheses.

(4) Collect passively; infer on-device where feasible; minimize retention; obtain meaningful household consent.

(5) Route high-stakes outputs to human/veterinary review; never override observable body signals.

(6) Treat the animal as a partner; keep welfare as the terminal criterion.

None of this slows the useful science; all of it protects the animals and the credibility of the field.

10 A Research Agenda: What Would Actually Move the Needle

If the distance estimate of Section 8 is right, then the highest-leverage work is not bigger models but better *infrastructure and protocols*. We propose a concrete agenda, ordered by leverage.

10.1 A public, grounded canine corpus

The field lacks what BEANS and BirdSet provide for general bioacoustics: a standardized, public, multimodal canine corpus with agreed labels [27]. The ideal corpus pairs synchronized audio, video (DogFACS-codeable face and posture), context metadata, and—crucially—*consequent behavior*, so that intent hypotheses can be checked. The soundboard citizen-science network, with thousands of instrumented homes, is the natural collection substrate [15]. Building this corpus with disclosed protocols and inter-annotator agreement would do more for the field than any single model release.

10.2 Calibrated, multimodal affect models

On the modeling side, the priority is fusion and calibration: combine bark, growl, face, and posture into a single state estimate with honest uncertainty, trained and evaluated on the public corpus. Self-supervised encoders (AVES, NatureLM-audio, Perch) supply the backbone [24, 25, 26]; the contribution is the canine-specific grounding and the calibration layer that makes the output safe for welfare use.

10.3 Closed-loop intent validation

The methodological frontier is the validation loop: infer an intent, act on it (or emit a candidate signal), and test the dog's response under blind conditions. The soundboard and CHAT protocols are the templates [16, 21]. A home-adapted, owner-blind protocol that scores intent inference against the dog's subsequent behavior would supply the error signal the field lacks (Section 7) and turn Rung 3 from hypothesis into measurement.

10.4 Structure discovery, with humility

Applying sequence-structure methods to bark corpora is worth doing—but with the explicit prior that canine displays are graded and may yield no clean combinatorial alphabet. The CETI and DolphinGemma toolkits are available [19, 21]; the discipline is to report structure as form, not meaning, and to heed the acoustic-similarity critique [30]. A negative result—no discrete inventory—would itself be valuable, sharpening the semantic-gap claim.

10.5 Engineered dialogue, not extracted dialogue

The most promising path to anything resembling conversation is *engineered*: expand the soundboard paradigm into a richer, negotiated symbol system, with the dog as partner and the behavior as check. This is Rung-3 work that may eventually look like dialogue, without pretending that spontaneous barks contain sentences. It is also the ethically preferable path (Section 9).

10.6 What would falsify our thesis

A good survey states what would change its mind. Our central claim—that Rungs 4–5 are ill-posed for natural canine communication—would be undermined by: (i) a reproducible demonstration of a productive, open referential vocabulary in spontaneous dog signals, validated by playback; or (ii) evidence of recursive, generative combination in natural canine displays (not trained soundboard sequences). Absent such evidence, the ladder stands. We invite the community to try to falsify it; that is how the distance estimate should shrink.

11 Conclusion

We began with a folkloric question—how far are we from translating what a dog says?—and end with a deflationary but, we hope, more useful answer. "Translation" is not one capability but five, and the distance to each is different in kind, not merely in degree. Signal detection and affect classification (Rungs 1–2) are here and shipping; validated affect-and-intent inference (Rung 3) is a research program with a foothold in the soundboard literature and a plausible horizon around 2030; open referential translation and free dialogue (Rungs 4–5) presuppose a generative symbolic system that the biological evidence does not support, and may be permanently ill-posed. The canine Rosetta Stone is, in the end, a metaphor with a catch: the original worked because one decree existed in three scripts, and for the dog there may be no decree to transcribe—only a rich, graded, affective life we can learn to read but not to render as sentences.

Three findings anchor this verdict. First, the biology: dogs signal affect, arousal, identity, motivation, and—with human scaffolding—referential intent, but they do not possess an open-ended generative grammar [1, 4, 6, 16]. Second, the technology: the decoding pipeline's front (encoding, structure discovery) is maturing fast on the back of the cross-species foundation-model boom, while its back (grounding, validation) is where canine-specific work is needed and where commercial effort is absent [24, 25, 19, 21]. Third, the evaluation: the field's headline accuracies are unfalsifiable as translation metrics, because no ground-truth intended message exists to score against; they are, at best, Rung-2 emotion-classification numbers [5]. The commercial market demonstrates Rung 2 and sells Rung 5, and the gap between those is the subject of this survey.

The way forward is not more scale but better infrastructure and protocols: a public grounded canine corpus, calibrated multimodal affect models, closed-loop blind validation, and an ethics framework that treats the animal as a partner and keeps welfare as the terminal criterion. The useful, achievable goal is *interpretation with calibrated uncertainty*, not transcription of a language that may not exist. That goal is within reach this decade, and it is worth reaching.

11.1 Limitations of this survey

In the spirit of evidence-bounded research, we state the limitations plainly. (1) This is a narrative survey, not a PRISMA systematic review; sources were gathered through structured web and literature search in November 2026, and coverage is broad rather than exhaustive. (2) Several fast-moving items (WhAM, ZF-AIM, vendor metrics) rest on project reports and journalism rather than peer review, and are labeled as such; commercial accuracies are reported *as claims* and were not independently verified. (3) Some citations were confirmed to the level of venue, year, and DOI via secondary indexes rather than full-text retrieval; a production version should verify each against the primary record. (4) The distance estimates are expert judgments from

the reviewed evidence, not forecasts, and Section 10 states what would falsify them. These limitations do not weaken the central claim—they bound it, which is the point.

Acknowledgments and tools

This survey was prepared with the assistance of open research-agent tooling, including the Scientific Agent Skills literature-review and autoresearch workflows [38], which contributed the search-documentation and citation-verification discipline used here. The figures are original TikZ constructions released with the repository for reuse and correction.

Data and code availability

The LaTeX source, both hero figures, the bibliography, and the project page are openly available in the `woof-to-word` repository: `https://github.com/mcxiaoxiao/woof2word`. Corrections and falsification attempts are welcome.

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